

Ansh Ankul

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AI Engineer and Forward Deployed Engineer building production LLM agents, evaluation systems, and multimodal AI pipelines. Experienced in translating ambiguous domain requirements into deployed AI products across enterprise integrations, benchmarking, and real-world data systems.

Work Experience

Lead AI Engineer, *National Center for Supercomputing Applications (NCSA)*, Champaign, Illinois. **Aug 2025 – Present**

- Lead technical development of **AI-AgriBench**, a public benchmarking platform evaluating 25+ LLMs and vision-language models across 6+ datasets; own benchmark architecture, evaluation infrastructure, and release workflows.
- Engineered scalable Python pipelines to ingest, deduplicate, semantically filter, and perform LLM-assisted quality control over 80,000+ agricultural QA pairs, producing curated evaluation and RAG datasets for **CropWizard**.
- Developed and deployed a full-stack annotation and evaluation platform used by agricultural experts to validate datasets, score model responses, and review benchmark results, using Next.js, React, FastAPI, and SQLite.
- Operationalize ambiguous evaluation requirements from researchers, domain experts, and external partners into production APIs, expert-review workflows, benchmark dashboards, and public leaderboard releases.

Forward Deployed Engineer (Part-time), *AgAnswers Inc.*, Champaign, Illinois. **May 2025 – Present**

- Designed and deployed tool-using LLM agents with n8n, translating natural-language requests into multi-step SQL-backed inventory and equipment-management workflows against live customer data.
- Integrated John Deere Operations Center APIs using OAuth 2.0 and recursive ingestion of nested equipment, location, operating-hour, and alert resources into Supabase/Postgres for live agent queries.
- Implemented mock farm datasets and local test fixtures to validate multi-step agent workflows without depending on live customer accounts, accelerating integration development and debugging.

Machine Learning Research Assistant, *University of Illinois Urbana-Champaign*, Champaign, Illinois. **Jan 2024 – Dec 2024**

- Trained and evaluated ML and neural models for predicting subsurface rock permeability from 10,000+ mineralogical records, achieving an R^2 of 0.93 on a held-out 20% test set.

Data Science Intern, *YAHOO!*, Remote / Champaign, Illinois. **Jun 2024 – Aug 2024**

- Built an anomaly-detection pipeline for advertising-revenue time series, evaluating multiple unsupervised approaches and improving recall by 25% over the prior baseline.
- Developed Prophet and ARIMA forecasting models with interactive Looker dashboards and automated alerting, improving RMSE by 20% and reducing false-positive alerts by 30%.
- Containerized and deployed model services on AWS using Docker and Kubernetes, reducing deployment time by 35% through CI/CD automation and reusable infrastructure templates.

Software Engineer, *BIGSTEP TECHNOLOGIES*, Gurugram, India. **Jan 2022 – Jun 2023**

- Engineered frontend features, backend APIs, and database integrations for a full-stack marketplace serving 1,000+ users using TypeScript, React/Next.js, Node.js, Flask, and MongoDB.
- Trained an NLP-based content-moderation pipeline using spaCy, NLTK, and RoBERTa to classify and moderate 1,000+ user-generated comments.

Data Science Intern, *ERNST & YOUNG*, Gurugram, India. **Jun 2021 – Aug 2021**

- Evaluated 15+ machine-learning approaches for anti-money-laundering detection on 17,000+ transactions and automated hyperparameter optimization with Optuna, improving F1-score from 0.82 to 0.88 while reducing tuning time by 40%.

Education

Master of Science in Statistics, *University of Illinois Urbana-Champaign*, Illinois, USA. **Aug 2023 – May 2025**

Bachelor of Technology in Computer and Communications Engineering, *Manipal University Jaipur*, Rajasthan, India. **Jul 2018 – Jul 2022**

Selected Projects

LLM Council Image Annotation Pipeline, *National Center for Supercomputing Applications*. **Aug 2025 – Present**

- Designed and integrated a multi-VLM annotation pipeline into AI-AgriBench using consensus voting and quality-control checks to generate reproducible labels across 10,000+ agricultural images.
- Evaluated consensus-generated labels against 500 expert-annotated images, achieving 91% agreement and analyzing failure modes across crop-disease, pest, and nutrient-deficiency categories.

Intelligent Document Retrieval System using RAG. **Feb 2024 – May 2024**

- Designed and evaluated a GPT-based RAG system combining FAISS with an entity graph across 5,000 queries, improving retrieval precision by 40% over a dense-vector baseline.
- Compared standard and graph-augmented retrieval, reducing irrelevant retrievals by 10% and improving grounding on multi-hop questions.

Technical Skills

Languages: Python, SQL, TypeScript/JavaScript, R, C++, Java

AI / ML: PyTorch, scikit-learn, Hugging Face, RAG, agentic AI, tool/function calling, multimodal AI

Evaluation: LLM evaluation, VLM evaluation, benchmarking, expert validation, consensus-based evaluation

LLM / Agent Frameworks: n8n, Google ADK, LangGraph, LangChain, OpenAI API, Anthropic, Gemini

Backend / Data: FastAPI, Node.js, Next.js, React, PostgreSQL, Supabase, SQLite, MongoDB, REST APIs

Infrastructure: AWS, Docker, Kubernetes, Git, Spark